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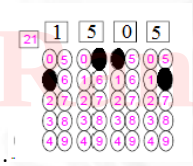
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 15-03-2026**

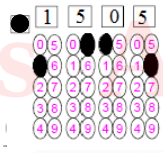
Time: 09.00AM to 12.00PM **GTM-22** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below





Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. For a unique of μ & λ the system of equations given by

$x+y+z=6$

$x+2y+3z=14$

$2x+5y+\lambda z=\mu$

Has infinitely many solutions, then $\frac{\mu-\lambda}{4}$ is equal to

1) 3 **2)** 5 **3)** 7 **4)** 9

2. Let $f(x)=\begin{cases}\frac{a(1-x\sin x)+b\cos x+5}{x^2}, & x<0\\ 3, & x=0.\\ \left[1+\left(\frac{cx+dx^3}{x^2}\right)\right]^{1/x}, & x>0\end{cases}$ If f is continuous at x=0, then $(a+b+c+d)$ is

1) 5 **2)** -5 **3)** $-5+\log_e 3$ **4)** $5-\log_e 3$

3. Let $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+.....+f\left(\frac{59}{15}\right)\right)$ is equal to

1) 118 **2)** 92 **3)** 102 **4)** 108

4. Let P be the set of seven-digit numbers with sum of their digits equal to 11. If the numbers in P are formed by using the digits 1, 2 and 3 only, then the number of elements in the set P is

1) 158 **2)** 173 **3)** 164 **4)** 161

5. The least value of n for which the number of integral terms in the binomial expansion of $\left(\sqrt[3]{7}+\sqrt[12]{11}\right)^n$ is 183, is k then the number of even divisors of k is

1) 24 **2)** 20 **3)** 2184 **4)** 32

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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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6. Let the points $\left(\frac{11}{2},\alpha\right)$ lie on or inside the triangle with sides $x+y=11, x+2y=16$ and $2x+3y=29$. The smallest and largest values of α are α_1,α_2 then the eccentricity of the ellipse whose length of major axis is α_2 and minor axis is α_1 is_____.

1) $\sqrt{\frac{265}{12}}$ **2)** $\sqrt{\frac{23}{12}}$ **3)** $\frac{\sqrt{23}}{12}$ **4)** $\frac{\sqrt{265}}{12}$

7. In a triangle ABC, BC=3 and AC =4 and circle with AB as diameter passes through the centroid of a triangle, then AB is

1) $\sqrt{5}$ **2)** 5 **3)** $\sqrt{3}$ **4)** $\frac{5}{2}$

8. Let f (x) be a cubic polynomial on R which increases in the interval $(-\infty,0)$ and in $(1,\infty)$ and decreases in the interval $(0,1)$. If $f'(2)=6$ and $f(2)=2$, Then the value of $\tan^{-1}(f(1))+\tan^{-1}\left(f\left(\frac{3}{2}\right)\right)+\tan^{-1}(f(0))$ is equal to:

1) $\tan^{-1}2$ **2)** $\cot^{-1}2$ **3)** $-\tan^{-1}2$ **4)** $-\cot^{-1}2$

9.

	Column I		Column II
(A)	A is a matrix such that $A^2=A$. If $(I+A)^8=I+\lambda A$, then $\lambda+1$ is equal to	(P)	64
(B)	If A is a square matrix of order 3 such that $\lvert A\rvert=2$, then $\left\lvert\left(adjA^{-1}\right)^{-1}\right\rvert$ is equal to	(Q)	1
(C)	Let $\lvert A\rvert=\lvert a_{ij}\rvert_{3\times3}\neq0$. Each element a_{ij} is multiplied by λ^{i-j}. Let $\lvert B\rvert$ the resulting determinant, where $\lvert A\rvert=\lambda\lvert B\rvert$, then λ is equal to	(R)	256
(D)	If A is diagonal matrix of order 3×3 is commutative with every square matrix of order 3×3 under multiplication and trace(A)=12, then $\lvert A\rvert=$	(S)	4

1) $A\to R, B\to S, C\to Q, D\to P$ **2)** $A\to P, B\to S, C\to Q, D\to R$

3) $A\to P, B\to S, C\to R, D\to Q$ **4)** $A\to R, B\to P, C\to Q, D\to S$

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10. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $\left|\hat{a}+\hat{b}+2\left(\hat{a}\times\hat{b}\right)\right|=2$, if $\theta\in(0,\pi)$ is the angle between $\hat{a}$ and $\hat{b}$. Then $\theta=$

1) $\frac{2\pi}{3}$ **2)** $\frac{\pi}{3}$ **3)** $\frac{\pi}{2}$ **4)** $\frac{\pi}{4}$

11. The general solution of the differential equation $\left(x-y^2\right)dx+y\left(5x+y^2\right)dy=0$ is

1) $(y+2x)^4=C\left|\left(y^2+x\right)^3\right|$ **2)** $\left|(y+2x)^3\right|=C\left(2y^2+x\right)^4$

3) $\left(y^2+x\right)^4=C\left|\left(y^2+2x\right)^3\right|$ **4)** $\left(y^2-x^4\right)=C\left|\left(y^2+2x\right)^3\right|$

12. The value of $\int_{\log\sqrt{\frac{\pi}{2}}}^{\log\sqrt{\pi}}\frac{e^{2x}}{\cos^2\left(\frac{1}{3}e^{2x}\right)}dx=$

1) $\sqrt{3}$ **2)** $\frac{1}{\sqrt{3}}$ **3)** $\frac{3\sqrt{3}}{2}$ **4)** $\frac{1}{2\sqrt{3}}$

13. For a group of 50 male workers, the mean and the standard deviation of their daily wages are Rs.630 and Rs.90 respectively and for a group of 40 female workers these are Rs.540 and Rs.60 respectively. Then, the standard deviation of all these 90 workers is

1) 60 **2)** 70 **3)** 80 **4)** 90

14. Let $P_1: y=3x^2$ *and* $P_2: y=x^2+18$ be two parabolas, if the area of the bounded region enclosed between P_1 & P_2 is 12 times the area of the region bounded between the line $y=kx(k>0)$ *and* P_1 *then* $4k^3$ is equal to_____.

1) 80 **2)** 81 **3)** 27 **4)** 1296

15. Let z be a complex number such that $\left|\frac{z+3i}{z-i}\right|=\frac{1}{2}, z\neq i$ then, z lies on the circle with centre (g,f) and radius 'r' then $|g+f-r|=$ _____

1) -7 **2)** 7 **3)** 8 **4)** -8

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16. Suppose that foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ are $(f_1,0)$ and $(f_2,0)$ where $f_1>0$ and $f_2<0$. Let p_1 and p_2 be two parabolas with common vertex at $(0,0)$ and foci at $(f_1,0)$ and $(2f_2,0)$ respectively. The distance between the directories of P_1 and P_2 is

1) 2 **2)** 4 **3)** 6 **4)** 8

17. If the domain of the function $f(x)=\frac{\sqrt{x^2-25}}{(4-x^2)}+\log_{10}(x^2+2x-15)$ is $(-\infty,\alpha)\cup[\beta,\infty)$ then $\alpha^3+\beta^2$ is equal to:

1) 100 **2)** -100 **3)** -150 **4)** 150

18. let $f:[a,b]\to[1,\infty)$ be a continuous function and let $g:R\to R$ be defined as

$$g(x)=\begin{cases}0 & if \quad x<a\\ \int_a^x f(t)dt & if \quad a\le x\le b\\ \int_a^b f(t)dt & if \quad x>b\end{cases}, \text{Then}$$

1) $g(x)$ is continuous but not differentiable at 'a'

2) $g(x)$ is differentiable on R

3) $g(x)$ is discontinuous at 'b'

4) $g(x)$ is continuous and differentiable at either 'a' or 'b' but not both.

19. The line passing through the extremity A of the major and extremity B of the minor axis of the ellipse $x^2+9y^2=9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin 'O' is

1) $\frac{31}{10}$ **2)** $\frac{29}{10}$ **3)** $\frac{21}{10}$ **4)** $\frac{27}{10}$

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20. Let $S=\{1,2,3,.....,2022\}$, then the probability, that a randomly chosen numbers n from the set S such that $\text{HCF}(n,2022)=1$ is

1) $\frac{128}{1011}$ **2)** $\frac{166}{1011}$ **3)** $\frac{127}{337}$ **4)** $\frac{112}{337}$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. How many ordered pair of integers (a,b) satisfy all the following inequalities

$a^2+b^2<16, a^2+b^2<8a, a^2+b^2<8b$

22. The roots of quadratic equation $3x^2-px+q=0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$ if the sum of the first 11 terms of this arithmetic progression is 88. Then $q-2p$ is equal to_____.

23. *Let* $\bar{c}$ *and* $\bar{d}$ be vectors such that $|\bar{c}+\bar{d}|=\sqrt{21}$ *and* $\bar{c}\times\left(4\hat{i}-\hat{j}+2\hat{k}\right)=\left(4\hat{i}-\hat{j}+2\hat{k}\right)\times\bar{d}$. *If* $\lambda_1,\lambda_2\,(\lambda_1>\lambda_2)$ are possible values of $|\bar{c}+\bar{d}|.(i-5j-4k)$ *then the equation* $(2\lambda_1-k)x^2+\left(k^2+5k-6\lambda_2\right)xy+k^2y^2-6x-8y-3\lambda_2=0$ represents a circle then '$-k$' equal to

24. Let P be the point (10,-2,-1) and Q be the foot of the perpendicular drawn from the point $R(1,7,6)$ on the line passing through the points $(2,-5,11)$ and $(-6,7,-5)$.then the length of the line segment PQ is equal to

25. Let $f(x)=\begin{cases} xe^{ax} & ,x\le 0 \\ x+ax^2-x^3 & ,x>0 \end{cases}$, where 'a' is a positive constant The interval in which $f'(x)$ is increasing is $\left[\frac{k}{a},\frac{a}{l}\right]$ Then $k+l$ is equal to

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. Match the List I with List II

List-I	List –II
A. Magnetic induction	I. $MLT^{-2}A^{-2}$
B. Magnetic flux	II. $ML^2T^{-2}A^{-2}$
C. Magnetic Permeability	III. $ML^0T^{-2}A^{-1}$
D. Self Inductance	IV. $ML^2T^{-2}A^{-1}$

Choose the correct answer from the option given below.

1) A-IV,B-III,C-I,D-II **2)** A-III,B-IV,C-II,D-I

3) A-I,B-III,C-IV,D-II **4)** A-III,B-IV,C-I,D-II

27. Statement-1

When two boats sail parallel in the same direction and close to each other, they are pulled towards each other.

Statement-2

The viscous drag on a spherical body is proportional to its speed

1) Both Statement-I and Starement-II are true

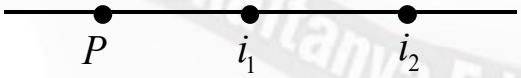
2) Statement-I is true but Statement-II is false

3) Both Statement-I and Statement-II are false

4) Statement-I is false but Statement-II is true

28. Assertion (A) :-

Figure shows cross-sections of two long straight wires; the left-hand wire carries current is directly out of the page. If the magnetic field due to the two currents is to be zero at P, the current i_2 must be into the page.



Reason (R) :- $i_2 > i_1$.

1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)

2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)

3) (A) is true but (R) is false

4) Both (A) and (R) are false

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29. Pick up the correct statements among the following in the aspect of an ideal gas.

I. For a given change in volume. Work done by gas during isothermal process is more than that of during adiabatic process.

II. Change in internal energy during adiabatic expansion is positive.

III. Work done by gas during isothermal process is given by $nRT\,ln\left(\frac{v_2}{v_1}\right)$

IV. Work done by the gas during free thermal expansion is zero.

V. Entropy is a path function

1) I,II are correct **2)** I,III,IV are correct

3) I,V are correct **4)** All statements are correct

30. Assertion (A): It is essential that the lines available in the emission spectrum will also be available in the absorption spectrum.

Reason (R): The spectrum of hydrogen atom is only absorption spectrum

1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)

2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)

3) (A) is true but (R) is false

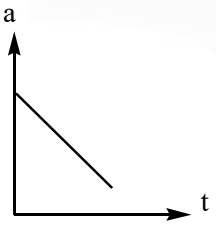
4) Both (A) and (R) are false

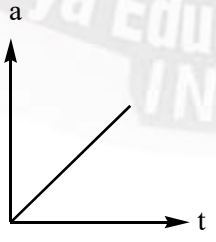
31. If an alpha particle with energy 7.7MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is …….m.

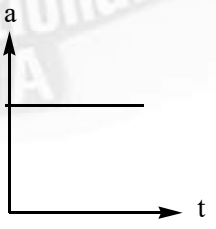
(Atomic number of gold = 79 and $\frac{1}{4\pi\in_0}=9\times10^9$ in SI units)

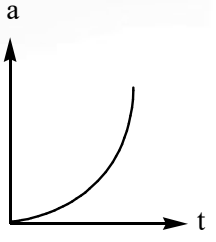
1) 2.95×10^{-14} **2)** 2.95×10^{-16} **3)** 3.85×10^{-16} **4)** 3.85×10^{-14}

32. The velocity v of an object varies with its position x on a straight line as $v=3\sqrt{x}$. Its acceleration versus time (a-t) graph is best represented by :

1) 

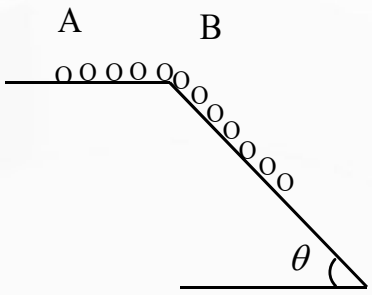
2) 

3) 

4) 

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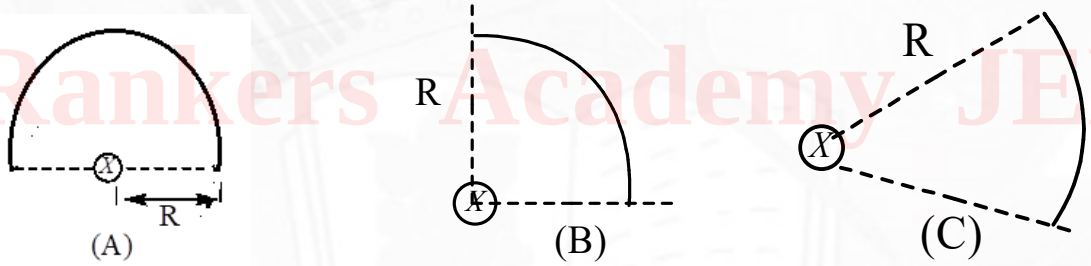
33. A chain of length L and mass m is places upon a smooth surface. The length of BA is (L-b). What will be the velocity of the chain when its end A reaches B?



1) $\sqrt{\frac{2g\sin\theta}{L}\left(L^2-b^2\right)}$ **2)** $\sqrt{\frac{g\sin\theta}{2L}\left(L^2-b^2\right)}$

3) $\sqrt{\frac{g\sin\theta}{L}\left(L^2-b^2\right)}$ **4)** None of these

34. In the three figure, each wire has a mass M, radius R and a uniform mass distribution. If they form part of a circle of radius R, then about an axis perpendicular to the plane and passing through the centre (shown by crosses), their moment of inertia in the order:



1) $I_A>I_B>I_C$ **2)** $I_A=I_B=I_C$ **3)** $I_A>I_B<I_C$ **4)** $I_A<I_C<I_B$

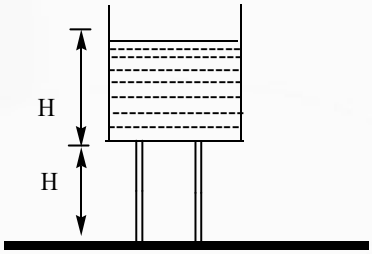
35. A rocket of mass M is launched vertically from the surface of the earth with an initial speed v. Assuming the radius of the earth to be R and negligible air resistance, the maximum height attained by the rocket above the surface of the earth is:

1) $\frac{R}{\left(\frac{gR}{2v^2}-1\right)}$ **2)** $R\left(\frac{gR}{2v^2}-1\right)$ **3)** $\frac{R}{\left(\frac{2gR}{v^2}-1\right)}$ **4)** $R\left(\frac{2gR}{v^2}-1\right)$

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36. A water tank is kept at a height H above the ground and the tank contains depth H of water as shown. Find the maximum possible range of water current, if there exists a small hole on the sidewall of the tank.



1) 2H **2)** $\frac{3H}{2}$ **3)** $\frac{5H}{2}$ **4)** H

37. The kinetic energy of translation of the molecules in 100g of CO_2 gas at 17^0C is (nearly)

1) 4205.5J **2)** 8217 J **3)** 6217 J **4)** 7217 J

38. An open organ pipe and a closed organ pipe vibrate at their first overtone frequencies, and the open pipe has twice the frequency of the closed pipe. If the close pipe has a length of 50 cm, find the length of the open pipe. (Take the speed of sound as 340 m/s)

1) 33 cm **2)** 112 cm **3)** 120 cm **4)** 132 cm

39. An electron of mass 'm' with an initial velocity $\hat{v} = v_0\hat{i}\,(v_0 > 0)$ enters an electric field $E = -E_0(\hat{i} + \hat{j})$.If the initial de Broglic wavelength is λ_0, the value after time t would be :

1) $\frac{\lambda_0}{\sqrt{\left(1+\frac{2E_0^2e^2t^2}{m^2V_0^2}+\frac{2E_0et}{mV_0}\right)}}$ **2)** $\frac{\lambda_0}{\sqrt{\left(1+\frac{E_0^2e^2t^2}{m^2V_0^2}+\frac{2E_0et}{mV_0}\right)}}$

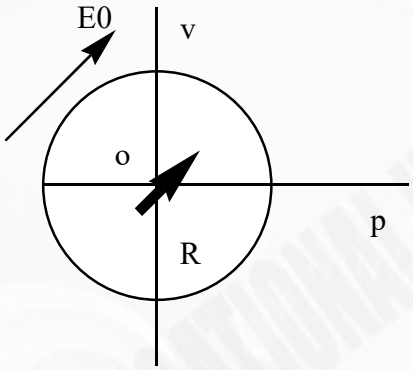
3) $\frac{\lambda_0}{\sqrt{\left(1+\frac{E_0^2e^2t^3}{m^2V_0^2}+\frac{E_0et}{mV_0}\right)}}$ **4)** $\frac{\lambda_0}{\sqrt{\left(1+\frac{2E_0^2e^2t^2}{m^2V_0^2}+\frac{E_0et}{mV_0}\right)}}$

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40. An electric dipole with dipole moment $\frac{P_0}{2}(\hat{i}+\hat{j})$ is held fixed at origin O in the presence of an external electric field of magnitude E_0. If the equipotential surface is a circle of radius R centered at origin as shown in the fig, then R is…….

(R >>> dipole size)



1) $R=\left[\frac{P_0}{4\pi\varepsilon_0 E_0}\right]^{1/3}$ **2)** $R=\left[\frac{2P_0}{4\pi\varepsilon_0 E_0}\right]^{1/3}$

3) $R=\left[\frac{P_0}{4\pi\varepsilon_0 E_0}\right]^{1/2}$ **4)** $R=\left[\frac{2P_0}{4\pi\varepsilon_0 E_0}\right]^{1/2}$

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41. The radiation emitted when an electron jumps from n=3 to n=2 orbit in hydrogen atom falls on a metal to produce photoelectrons. If the ejected photo electrons with maximum kinetic energy move in circular path of radius 0.1 cm in transverse magnetic filed of $B=\frac{1}{320}T$, then work function of metal is nearly

1) 2 eV **2)** 1 eV **3)** 0.5 eV **4)** 1.5 eV

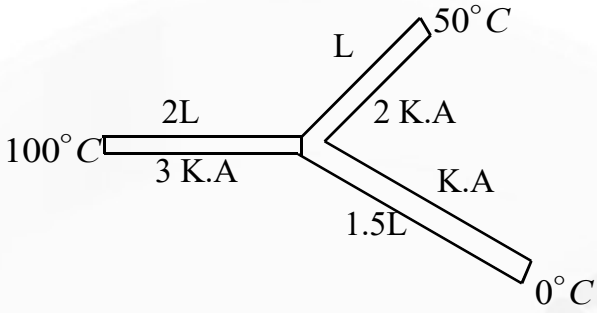
42. A solenoid has 2000 turns wound over a length of 3.0 m. The area of cross-section is $1.2\times10^{-3}m^2$. Around its central section a coil of 3000 turns is closely wound. If an initial current of 2A is reversed in 0.25 s, find the emf induced in the coil

1) 48 mv **2)** 25 mv **3)** 36 mv **4)** 43 mv

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43. Three rods are joined and free ends are maintained at constant temperature as shown in figure, steady state rate of heat flow through the rod of smallest length is

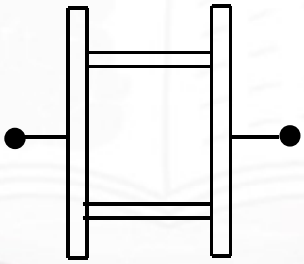


1) $\frac{40KA}{L}$ **2)** $\frac{80KA}{3L}$ **3)** $\frac{30KA}{L}$ **4)** $\frac{20KA}{L}$

44. In normal adjustment, for an astronomical telescope, the distance between objective and eye piece is 30cm. The focal length of the objective, when the angular magnification of the telescope is 2, will be :

1) 20 cm **2)** 30 cm **3)** 10 cm **4)** 5 cm

45. A capacitor is to be designed to be operated with constant capacitance, in environment of variable temperature. Parallel plate capacitor contains a 'space' to change the distance for compensation of temperature effect. If α_1 and α_2 be the coefficients of linear expansion of plates and space respectively and there is no change in capacitance with change of temperature, then relation between α_1 and α_2 is



1) $3\alpha_1 = \alpha_2$ **2)** $\alpha_1 = 2\alpha_2$ **3)** $\alpha_1 = 3\alpha_2$ **4)** $2\alpha_1 = \alpha_2$

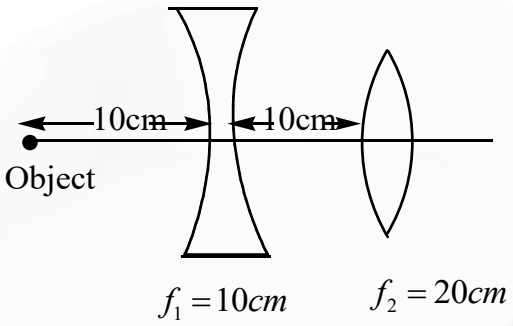
SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

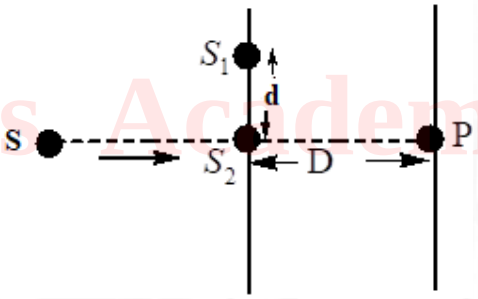
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

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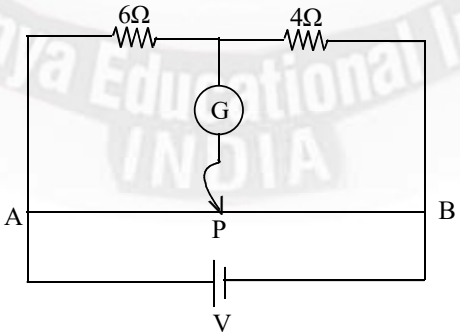
46. In the arrangement shown, distance of final image from convex lens (in cm) in magnitude is _____.



47. In a YDSE experiment two slits S_1 and S_2 have separation of d = 2 mm. The distance of the screen is D = 8/5 m. Source S starts moving from a very large distance towards S_2 perpendicular to S_1S_2 as shown in figure. The wavelength of monochromatic light is 500 mm. The number of maximas observed on the screen at points P as the source moves towards S_2 is 3995 + n. The value of n is…..

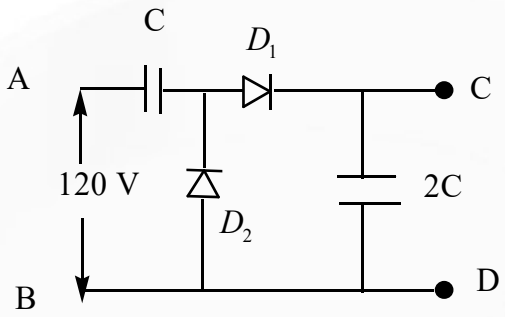


48. The total length of potentiometer wire AB is 50cm in the arrangement shown in figure. If P is the point where the galvanometer shows zero reading ,then the length AP is….cm..

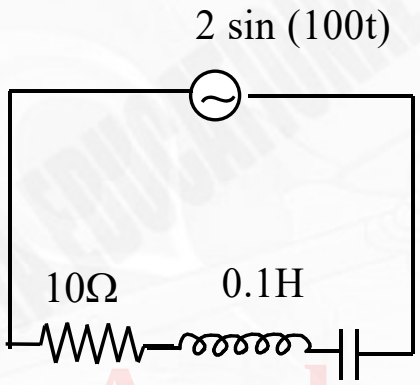


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49. D_1 and D_2 are ideal diodes . A 120 v battery is connected to the input terminals with A at positive and B at lower potential. Find output voltage (in volt) across C and D.



50. The power factor of circuit is $1/\sqrt{2}$. The capacitance of the capacitor is equal to_____ μF



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CHEMISTRY

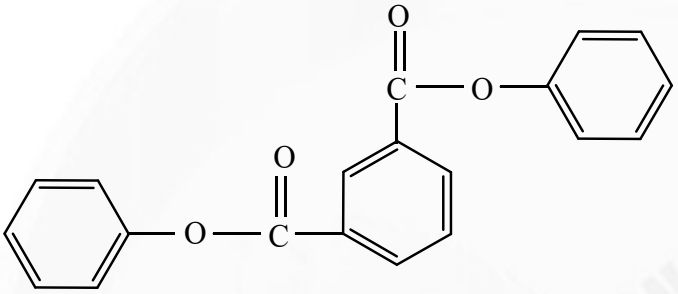
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

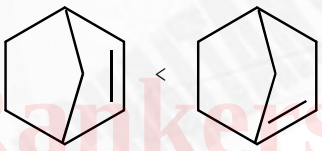
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

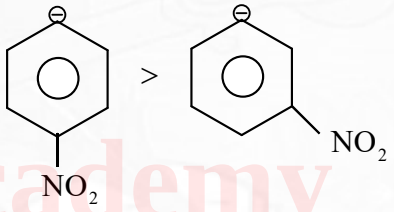
51. The correct IUPAC name of the compound:



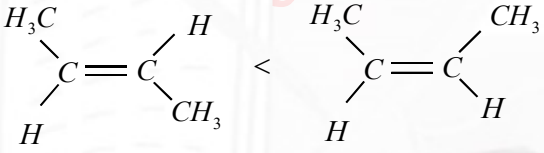
1) Phenyl-3-phenoxycarbonylbenzoate
2) Phenyl-3-phenoxycarbonylbenzenecarboxylate
3) Diphenylbenzene-1,3-dicarboxylate
4) 1,3-Diphenoxycarbonylbenzene

52. Correct order regarding stability is

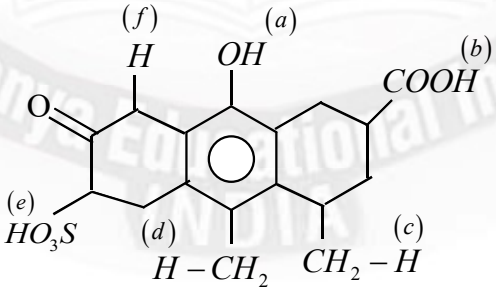
1) 

2) 

3) $\overset{\ominus}{C}F_3 < \overset{\ominus}{C}Cl_3$

4) 

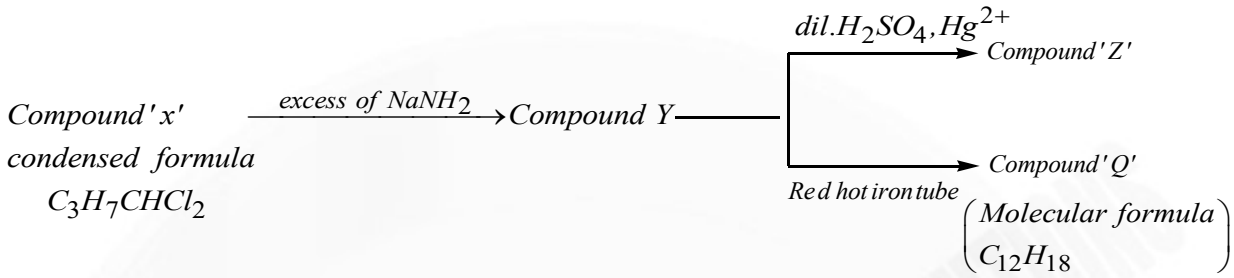
53. Arrange the following in increasing order of acid strength.



1) $c > f < d < a < b < e$
2) $f < d < c < b < e < a$
3) $f < c < d < b < e < a$
4) $c < d < f < a < b < e$

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54. Given below are two statements for the following reaction sequence.



Statement I : Compound Z will give yellow precipitate with NaOI.

Statement II: Compound 'Q' has two different types of 'H' atoms (aromatic : aliphatic) in the ratio 1 : 5

In the light of the above statements choose the correct answer from the options

1) Statement I is true but Statement II is false

2) Both statement I and statement II are false

3) Statement I is false but statement II is true.

4) Both Statement I and statement II are true.

55. Consider the following reaction sequence.

$$Compound\,(x) \xrightarrow[(ii)H_3O^+]{(i)CO_2, NaOH, 120^0C,\ high\ pressure} Compound\,(y)$$

$[76.6\%C, 6.38\%H,$ vapour density 47$]$ (major product)

Compound (y) develops characteristic colour with neutral $FeCl_3$ solution.

Identify the INCORRECT statement from the following for the above sentence.

1) Both compound x and y will dissolve in NaOH

2) Compound y will dissolve in $NaHCO_3$ and evolve a gas.

3) Both compound x and y will burn with sooty flame.

4) Compound x is more acidic than compound y

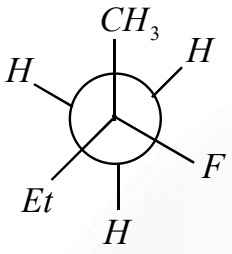
Choose The Correct Answer

1) 1,2 **2)** 2,3 **3)** 1,2,3 **4)** Only 4

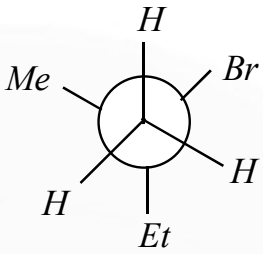
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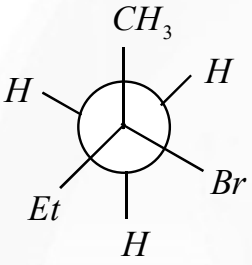
56. In which of the following cases Hofmann's product is the major product?

P. 

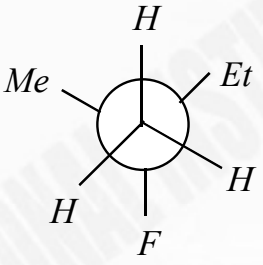
$\xrightarrow[\Delta]{alc.KOH}$

Q. 

$\xrightarrow[(ii)\ Moist\ Ag_2O/\Delta]{(i)\ Et_3N}$

R. 

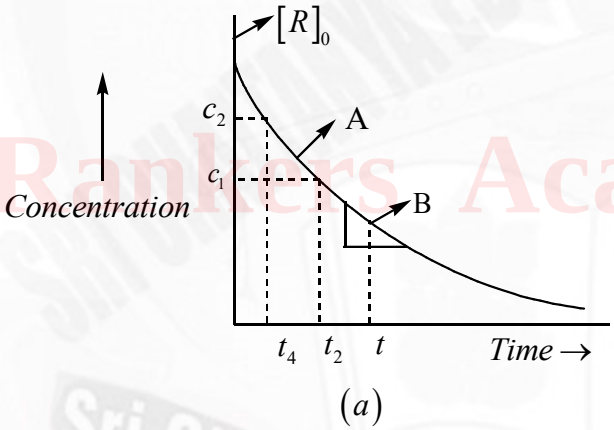
$\xrightarrow[\Delta]{pot.-t-butoxide}$

S. 

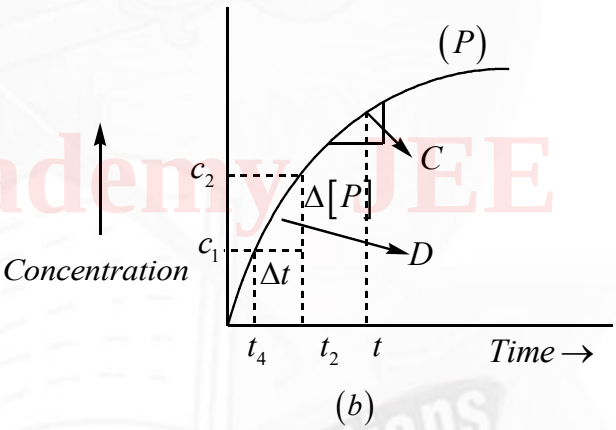
$\xrightarrow[\Delta]{pot.-t-butoxide}$

1) Only P **2)** Only Q & R **3)** All P, Q, R & S **4)** Only S

57. Study the following graphs and choose the correct option



(a)



(b)

i) in fig.a A represents average rate and B represents instantaneous rate

ii) in fig.b D represents average rate and C represents instantaneous rate

iii) fig. a, A represents instantaneous rate and B represents average rate

iv) fig. b, C represents average rate and D represents instantaneous rate

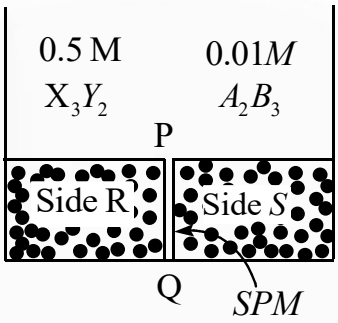
1) i and ii are correct **2)** ii and iv are correct

3) i and iv are correct **4)** ii and iii are correct

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58. $X_3Y_2\,(i=5)$ when reacted with $A_2B_3\,(i=5)$ in aqueous solution, gives brown colour. These are separated by a semipermeable membrane PQ as shown. Due to osmosis there is:



1) brown colour formation in side R **2)** brown colour formation in side S

3) formation in both of the sides R and S **4)** no brown colour formation

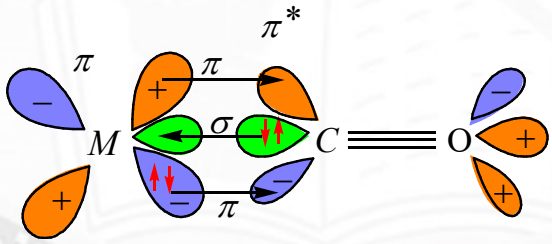
59. The equilibrium constant for the reaction is:

$In^{2+} + Cu^{2+} \rightarrow In^{3+} + Cu^{+}$ *at* $298K$

Given, $E^{\circ}_{Cu^{2+}/Cu^{+}} = 0.15V$, $E^{\circ}_{In^{3+}/In^{+}} = -0.42V$, $E^{\circ}_{In^{2+}/In^{+}} = -0.40V$

1) 10^5 **2)** 10^3 **3)** 10^{10} **4)** 10^6

60. Assertion (A): π – back bonding strengthen the $M-C$ bond order and weaken CO bond order



Synergic bonding

Reason (R) :

1) Both A and R are incorrect

2) A is correct but R is incorrect

3) Both A and R are correct but R is not the correct explanation of A

4) Both A and R are correct and R is the correct explanation of A

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61. Consider the following two complex ions: $[CoF_6]^{3-}$ *and* $[Co(C_2O_4)_3]^{3-}$. Which of the following statement(s) is/are false?

I. Both are octahedral.

II. $[Co(C_2O_4)_3]^{3-}$ is diamagnetic while $[CoF_6]^{3-}$ is paramagnetic.

III. Both are outer orbital complexes.

IV. In both the complexes the central metal is in the same oxidation state.

1) II and III only **2)** II, III and IV only

3) III only **4)** III and IV only

62. Which of the following option is correct about the following oxides?

$GeO, CO_2, CO, GeO_2, SiO_2, SnO_2, SnO$

1) $Acidic - GeO_2, CO, SiO_2$; $Neutral - GeO$
$Amphoteric - CO_2, SnO, SnO_2$

2) $Acidic - SnO_2, GeO_2, CO_2$; $Neutral - CO, SnO, SiO_2$
$Amphoteric - GeO$

3) $Acidic - CO_2, GeO_2, SiO_2$; $Neutral - CO$
$Amphoteric - SnO_2, SnO$

4) $Acidic - GeO, SiO_2, SnO$; $Neutral - CO, CO_2$
$Amphoteric - SnO_2, GeO_2$

63. Identify A to D, given that

$$A(\text{colourless solid}) \xrightarrow{HCl} Gas\ B(brown)$$
$$A(\text{colourless solid}) \xrightarrow{Zn/OH^-} Gas\ C(alkaline)$$
$$A(\text{colourless solid}) \xrightarrow{\Delta} Gas\ D(less\ reactive)$$

i) NH_4NO_3 *ii*) NH_4NO_2 *iii*) $NaNO_2$ *iv*) NH_3

v) NO *vi*) N_2 *vii*) HNO_3 *viii*) NO_2

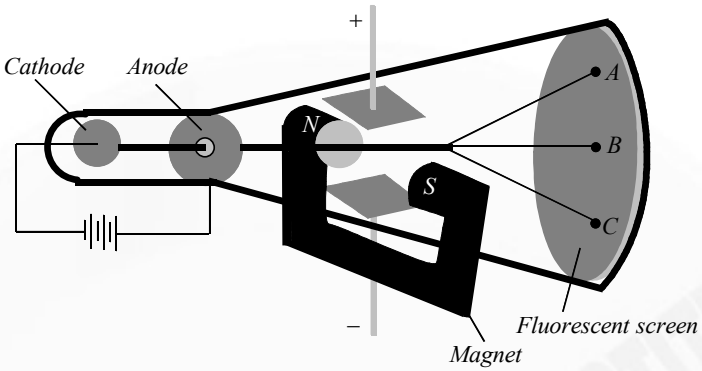
P. Gas B is viii Q. Solid A is ii R. Gas C is iv S. Gas D is v

Correct combination is

1) Only P **2)** P, Q, R **3)** P & S **4)** Only R

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64. On the basis of figure given below which of the following statement(s) is/are correct?



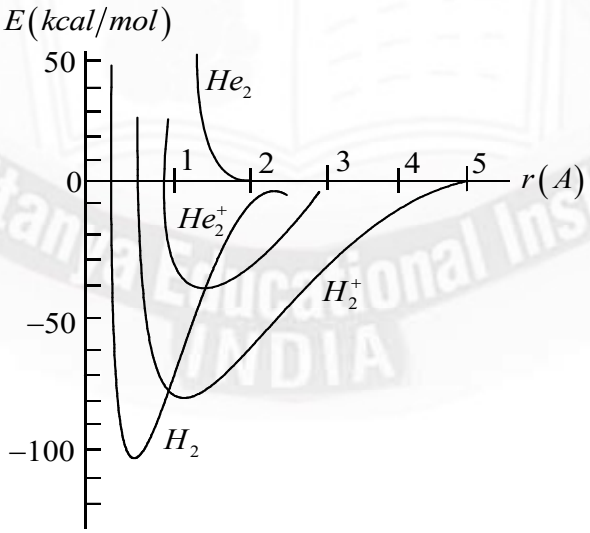
Electron strikes:
(i) At point B, when only electric field is applied.
(ii) At point C, when both electric and magnetic field is applied.
(iii) At point B, when both electric and magnetic fields are balanced.
(iv) At point C, when only magnetic field is applied.

1) (i) and (ii) **2)** only (iii) **3)** (iii) and (iv) **4)** (i) and (iii)

65. The heat of combustion of ethane gas is $368\ kcal/mol$. Assuming that 60% of the heat is useful, how many m^3 of ethane measured at STP must be burnt to supply enough heat to convert $50kg$ of water at $10°C$ to steam at $100°C$? $[STP = 22.4L]$
(Specific heat of water is $1\ cal/g$, heat of vaporization of H_2O is $540\ cal/g$)

1) $2.196\ m^3$ **2)** $1.196\ m^3$ **3)** $4.196\ m^3$ **4)** $3.196\ m^3$

66. The following graph is given between total energy and distance between the two nuclei for species H_2^+, H_2, He_2^+ *and* He_2 which of the following statement is correct?



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1) He_2^+ is more stable than H_2^+

2) Bond dissociation energy of H_2^+ is more than bond dissociation energy of He_2^+

3) Since, bond orders of He_2^+ *and* H_2^+ are equal hence both will have equal bond dissociation energy

4) Bond length of H_2^+ is less than bond length of H_2

67. Match the pair of salt given in column-I with the reagents used to distinguish them given in column-II

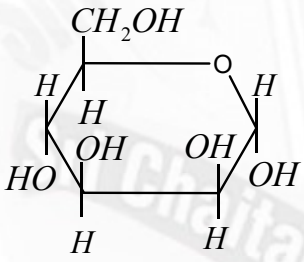
	Column-I		Column-II
1)	$FeSO_4$ & $FeCl_3$	P)	$NaOH$
2)	$MnSO_4$ & $ZnSO_4$	Q)	NH_4OH
3)	$CuSO_4$ & $FeSO_4$	R)	$K_4\left[Fe(CN)_6\right]$
4)	$Al_2(SO_4)_3$ & $ZnSO_4$	S)	PbO_2 & $conc.HNO_3$

1) $1-PQRS;2-PQR;3-QR;4-PQR$ **2)** $1-PQR;2-PRS;3-PQR;4-PS$

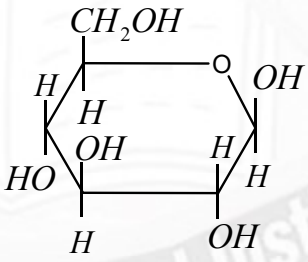
3) $1-PQR;2-PQRS;3-PQR;4-QR$ **4)** $1-PQRS;2-PQS;3-PQRS;4-PR$

68. D-Mannose differs from D-glucose in its stereochemistry at C-2. Which of the following is the correct structure of pyranose form of $\alpha-D(+)$ Mannose.

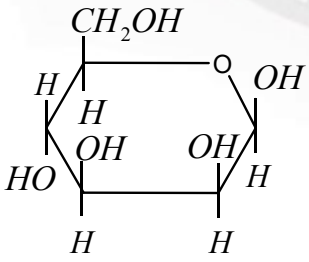
1)



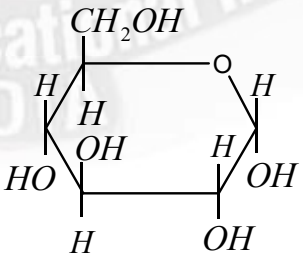
2)



3)



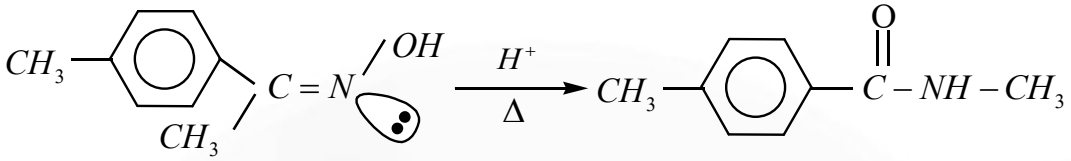
4)



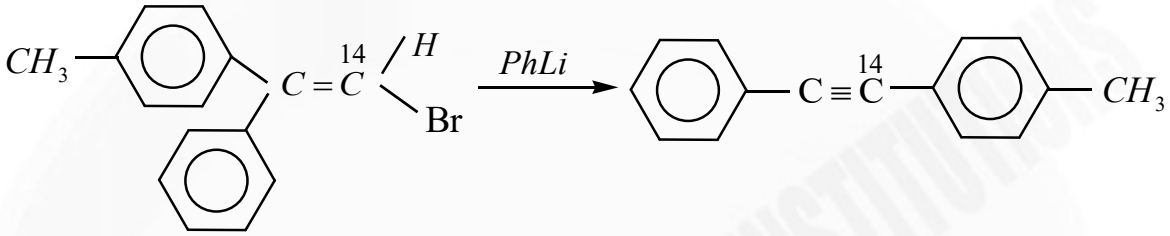
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69. Which of the following does not represent major product

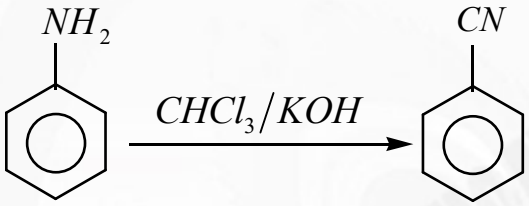
1)



2)



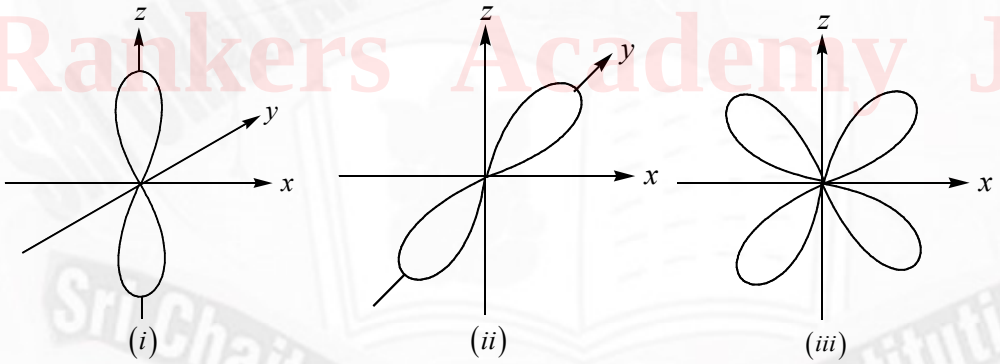
3)



4)



70. Consider the following three orbitals:



Correct statement(s) regarding given information is/are

P. Orbitals (i) and (ii) can never form any type of covalent bond between them

Q. If internuclear axis is x, then combination of (ii) and (iii) orbitals can form π – bond

R. Orbital (iii) can form δ – bond with orbital (iii) along the y-axis

S. If internuclear axis is x, then combination of (i) and (ii) orbitals can form π – bond along the y-axis

1) Only P & R **2)** Only Q & R **3)** Only R & S **4)** Q, R, S

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

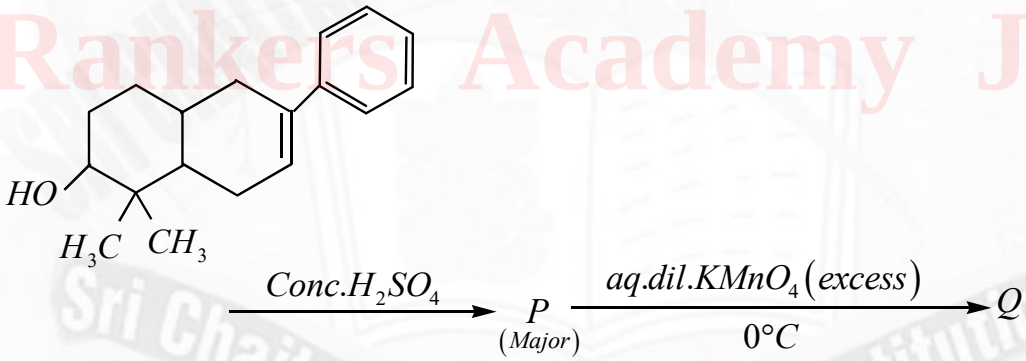
71. $A \rightarrow B(first\ reaction)$

$C \rightarrow D(Second\ reaction)$

Consider the above two first-order reactions. The rate constant for first reaction at 500 K is double of the same at 300K. At 500 K, 50% of the reaction becomes complete in 2 hour. The activation energy of the second reaction is half of that of first reaction. If the rate constant at 500 K of the second reaction becomes double of the rate constant of first reaction at the temperature; then rate constant for the second reaction at 300 K is _____ $\times 10^{-1} hour^{-1}$ (nearest integer).

72. Find the number of moles of $K_2Cr_2O_7$ reduced by three mole of Sn^{2+} ions.

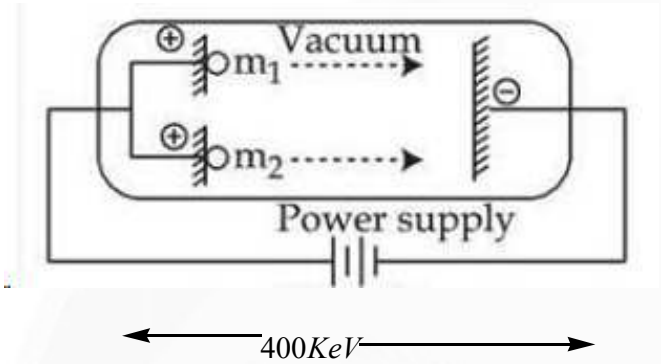
73. The number of hydroxyl group(s) in 'Q' is



74. At $27^0 C$, a solution containing 0.2 g of polyisobutylene in 100 ml of benzene developed a rise of 2.463 mm at osmotic equilibrium. The molar mass of polyisobutylene is $X \times 10^5\ g/mol$. Then X *is* _____ (Given: The density of final solution is 1.013 g/ml. $g = 10 m/s^2$ $R = 0.0821\ L-atm/K-mol$)

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75. Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 400 keV as shown below.



(Given mass of $m_1 = 4amu$ and $m_2 = 16amu$)

The debroglie wavelength of m_1 will be x times of m_2.The value of x is ……

(nearest integer)

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